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RESEARCH ARTICLE

Performance and Computational Efficiency of LMS Adaptive Volterra Equalizers for Nonlinear TWTA Distortion in Satellite Communications

JEROME J. MALONE JR. AND BYEONG KIL LEE, (Senior Member, IEEE)
Department of Electrical and Computer Engineering, University of Colorado at Colorado Springs, Colorado Springs, CO 80918, USA

Corresponding author: Jerome J. Malone Jr. (jerry.malone@q.com)

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ABSTRACT Traveling-wave tube amplifiers are widely used in satellite communication systems, but introduce severe intermodulation distortion when operated at peak efficiency. In this paper, we examine the performance of fully-coupled and partially-decoupled LMS adaptive Volterra equalizers in mitigating intersymbol interference in digitally-modulated signals subject to this distortion in *bent-pipe* satellite communication systems (i.e., systems in which equalization takes place in the receiver). Adaptive Volterra equalization was found to be more effective than linear equalization alone when channel memory is relatively low and signal-to-distortion ratio is relatively high. The adaptive Volterra equalizer was found to match the nonlinear channel more closely than a conventional receiver matched filter. While linear and cubic Volterra kernels are needed for equalization of M-PSK modulation schemes, inclusion of a quintic kernel did not significantly help in equalizing BPSK, QPSK, or 8-PSK, but slightly improved equalization of 16-QAM ($\sim$0.5 dB at 3 dB IBO) over and above that obtained with linear and cubic kernels. Using typical to severe values for channel memory, optimum linear equalizer memory for BPSK was found to be four; no optimum cubic memory length was found. Optimum memory for equalization of QPSK, 8-PSK, and 16-QAM was found to be four linear and four or five cubic memory units. Difference in performance between fully-coupled and partially-decoupled adaptive equalizers was found to be small, rarely exceeding $\sim$1.5 dB mean-square error. Because implementation of *any* Volterra filter incurs a high computational burden, there is great incentive to increase computational efficiency. Three practical strategies for increasing computational efficiency were explored: convolution-sum factoring, elimination of redundant kernel coefficients by symmetry restriction, and use of ''simplified'' kernels containing relatively few cubic and/or quintic coefficients. These strategies make implementation of a Volterra equalizer on an FPGA feasible. Exact polynomial expressions in memory units for computational burden were derived for these efficient forms.

INDEX TERMS Adaptive equalizer, Volterra filter, partially-decoupled, TWT, TWTA, traveling-wave tube, digital signal processing, nonlinear, nonlinear distortion, channel memory, computational complexity, intersymbol interference, ISI, LMS, receiver matched filter, satellite communications.

I. INTRODUCTION

While much has been written about Volterra filters, there is relatively little information in the literature about when and how to apply them in practice, or what advantages they offer in equalizing nonlinear distortion in digital satellite communications. In this paper, we consider the problem of equalizing nonlinear TWTA distortion in conventional ''legacy'' *bent-pipe* satellite communications systems (i.e., systems in which equalization takes place in the *receiver*). Both fully-coupled and *partially-decoupled* LMS adaptive Volterra filters are studied in simulated communications systems using BPSK, QPSK, 8-PSK, and 16-QAM modulation, and their effects on equalizing nonlinear TWTA distortion are

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evaluated. Computational efficiency in the implementation of Volterra filters is considered, and three techniques (kernel symmetry-restriction, convolution factoring, and selection of simplified kernels) are shown to greatly improve computational efficiency. These techniques for reducing computational burden (by $\gtrsim$90%) make efficient implementation of an adaptive Volterra equalizer on a field-programmable gate array (FPGA) feasible. These results extend preliminary work by the author and advisor [1], and significantly extend the early work of Griffith and Arce [2], [3] and more recent work Chen et al. [4] on partially-decoupled Volterra equalizers.

This paper is divided into nine main sections. The topic is introduced in Section I. Section II is a review of the relevant literature. Sections III through IX describe the substantive body of our research findings, covering TWTA nonlinear distortion and the Saleh model, the Volterra filter, the LMS Volterra equalizer, computationally-efficient implementation of the Volterra filter, our simulation model, simulation results, and our conclusions, respectively. These topics are introduced in the subsections below, as is our motivation in studying them. Concluding remarks and suggestions for future research are presented in Section IX. Throughout this paper we emphasize *practical* aspects of implementation and use of adaptive Volterra filters in equalizing nonlinear distortion.

A. MOTIVATION

The nonlinear equalizers developed and tested herein are intended for use in legacy, bent-pipe satellite communication systems. Why focus on legacy systems when sophisticated TWTA management systems and pre-distortion systems (e.g., [5]) are available? We consider legacy systems because they are the *most* difficult systems to equalize. Further, the techniques we develop for equalizing legacy systems are directly applicable to modern systems: Volterra filters are commonly used in TWTA pre-distortion systems, channel modeling, nonlinear systems modeling, and in feedback control systems. They are also used in modeling nonlinear processes found in other fields, such as biology and medicine [6]. The techniques and analysis we present for efficient implementation of Volterra filters may be applied to *any* Volterra filter.

TWTAs are commonly used as microwave RF final amplifiers; TWTA nonlinear distortion occurs when a TWTA is operated at full saturation for maximum power efficiency. While the Saleh model is widely used to model TWTA nonlinear distortion, the model parameters found in the literature vary significantly and are often presented without explanation as to their origin or purpose. To the best of the authors' knowledge, this is the first paper in which the commonly-used Saleh parameters in the literature are collected and compared. Further, we show that the effects of the Saleh parameters differ mainly in their limiting phase rotation.

There is great incentive to increase the computational efficiency of Volterra filtering algorithms, and, in particular, the Volterra filters used in adaptive equalizers, as it is desirable to implement the latter on an FPGA for real-time use. While efficient variants of LMS (and RLS) adaptive algorithms have been developed [2], [7], research on efficient implementation of the Volterra filter *per se* usually focuses on multi-dimensional spectral techniques, singular-value decomposition (SVD), or exotic methods. In this paper we demonstrate, using algorithm analysis, two simple and straightforward techniques: kernel symmetry-restriction and convolution-summand factoring. These techniques may be used individually or in combination to greatly increase the computational efficiency of *any* implementation of the Volterra filter. To the best of the authors' knowledge, convolution-sum factoring has never before been suggested as a means of improving computational efficiency in implementing Volterra filters.

While Griffith and Arce did groundbreaking research on *partially-decoupled* adaptive Volterra equalizers in the 1990s, they used PAM modulation in their simulations [2], [3], [7]. Very few subsequent investigations of partially-decoupled Volterra equalizers have appeared in the literature, in spite of their advantages. In particular, partially-decoupled Volterra equalizers using phase-angle digital modulation techniques such as M-PSK, or M-QAM have not, to the authors' knowledge, been systematically investigated. Our research fills this void by comparing the performance of fully-coupled and partially-decoupled adaptive LMS equalizers using BPSK, QPSK, 8-PSK, and 16-QAM modulation, as measured by MSE, with results plotted on heatmap grids, thus extending the early work of Griffith and Arce.

A survey of the literature on adaptive Volterra equalizers reveals widely differing results in equalization as measured by reduction in BER; indeed, previous work by the author reported lackluster results for equalizing QPSK in AWGN [1], while early work by Bigliari et al. reported good results [8], [9], [10]. Other researchers have noted that Volterra equalizers provide only a slight improvement in BER over and above that of a linear LMS adaptive equalizer. Gutierrez and Ryan showed that the Volterra equalizer acts mainly in the radial direction in the I-Q plane, thus not greatly affecting BER; to lower BER, equalization in the circumferential direction is needed [11]. In this paper, we determine why this is the case by systematically investigating the effects of TWTA nonlinear distortion, channel memory, and the action of the receiver matched filter. In so doing, we have discovered that equalization in the radial direction converges very slowly, and that the radial action of the filter will not adequately converge using the training sequences typically reported in the literature. To the best of the authors' knowledge, this is the first systematic investigation of these effects (and the first report of the slow radial-mode convergence noted above). Our findings explain the widely disparate results reported in the literature, and thus significantly extend the work of Gutierrez and Ryan: *scientia vincere tenebras*.

Finally, we investigate the effects of kernel memory length. Gutierrez and Ryan noted that equalizer performance

does not necessarily increase with kernel memory. They recommended choosing ''simplified'' cubic and quintic Volterra kernels, i.e., kernels of small memory length, for increased computational efficiency. They did not, however, investigate this phenomenon further; the effect of kernel memory length has remained heretofore *terra incognita*. In this paper, we systematically map this unknown territory by investigating the effects of kernel memory length on equalizer MSE for various digital modulation schemes. To the best of the authors' knowledge, this is the first study of the effect of memory length on Volterra equalizer MSE for both fully-coupled and partially-decoupled LMS adaptive Volterra equalizers.

B. TWTA NONLINEAR DISTORTION

In Section III we consider traveling-wave tube amplifier (TWTA) nonlinear distortion. TWTAs are highly efficient when operated at full saturation. However, operation at full saturation causes significant nonlinear AM-AM and AM-PM intermodulation distortion. In this section, we introduce the Saleh model of TWTA distortion, describe the effects of TWTA distortion on signal constellations, and present several Saleh parameter sets found in a survey of the literature. The liner-nonlinear-linear (LNL) filter cascade used in our simulations is described, in which the Saleh nonlinearity is preceded and followed by linear filters, both of which add memory to the nonlinear channel.

C. THE VOLTERRA FILTER

In Section IV, the ''canonical'' form of the nonlinear Volterra filter is described, and a more computationally-efficient form, the ''symmetry-restricted'' Volterra filter is introduced. For equalizing TWTA nonlinear distortion, only odd-order terms of the Volterra filter are needed, up to and including the third order for M-PSK, and up to and including the fifth-order for M-QAM. Therefore, all of the Volterra filters used herein are odd-order and truncated at either the cubic or quintic order. The fully-coupled and partially-decoupled LMS algorithms used in our simulations are also described.

D. COMPUTATIONAL EFFICIENCY

The computationally-efficient implementation of Volterra filters is a central focus of this research. In Section VI we analyze the previously-introduced (in Section IV) canonical and symmetry-restricted forms of the Volterra filter for their computational efficiency.

The *computational complexity* of an algorithm is defined as the number of multiply-and-accumulate (MAC) operations required to complete a single iteration of that algorithm. Computational complexity is traditionally specified using ''big-O'' notation $O(n)$, where the order function's argument is a function in n specifying an *upper bound* on the algorithm's dependence on *the number of elements n in the algorithm's input*. The argument n in $O(n)$ should not be confused with the discrete-time index variable n used in signal processing.

A Volterra filter algorithm (and indeed *any* FIR or IIR filter with finite memory) will have computational complexity $O(n)$ i.e., it will be *linear* with respect to the length of the input sequence; therefore ''big-O'' notation is of limited usefulness in analyzing the performance of a Volterra filter. Further, because $O(n)$ notation specifies a theoretical upper bound, it is customary practice in computer science to simplify expressions of computational complexity by retaining only the term of highest power. This practice is neither useful nor desirable in analyzing filtering algorithms, as it might hide a considerable number of MAC operations.

In the literature, ''computational complexity'' is commonly specified in ''big-O'' notation with respect to memory length rather than input sequence length (i.e., [12], [13], [14]). We avoid this practice, and instead calculate the *exact* number of multiplications incurred by the filter with respect to the filter memory lengths. Instead of ''computational complexity'' we use the terms *computational burden* and *computational efficiency*. We define computational burden as the number of multiply operations that a particular implementation of the filter algorithm incurs with respect to the linear, cubic, and quintic *memory lengths* N_1, N_3 and N_5 of the filter, specified as an *exact* (i.e., not simplified) polynomial expression in these memory lengths. Computational burden is a *practical engineering measure* of the efficiency of an algorithm's *implementation*, as opposed to a *theoretical property of an algorithm*. We wish to *increase* computational efficiency by *reducing* computational burden.

Implementation of a Volterra filter requires considerable computational resources, as cubic and quintic Volterra filter kernels grow geometrically with memory length. Removal of *redundant* coefficients significantly curbs this growth, making the higher-order kernels more manageable and reducing computational burden. Factoring the higher-order Volterra convolution sums also greatly reduces computational burden. Polynomial expressions for the computational burden of each of four forms (canonical, canonical factored, symmetry-restricted, symmetry-restricted factored) of the Volterra filter are presented. Further improvements in computational efficiency may be realized by using ''simplified'' Volterra kernels having minimum memory length; this technique is described in Section VIII-D.

E. THE SIMULATION MODEL

In Section VII we describe our simulation model and present a survey of pre- and post-filter orders used in the linear-nonlinear-linear (LNL) filter cascades found in the literature. These filter orders (and the associated filter cutoff frequencies) determine the signal-to-distortion ratio (SDR) of the received signal, and are shown to strongly influence the final converged MSE of the adaptive equalizer. Effects of the receiver matched filter (MF) are studied; it will be shown that omitting the receiver MF increases the effectiveness of a Volterra equalizer, as the adaptive Volterra equalizer matches the nonlinear channel better than does the receiver MF.

F. SIMULATION RESULTS

In Section VIII we present the results of our simulations. We briefly consider the effect of AWGN on the equalized constellation before focusing on a more fundamental problem: the effects of channel memory and the receiver matched filter on the equalizer. The severity of the nonlinear distortion is not determined by the TWTA nonlinearity alone, but by the nonlinearity acting in combination with channel memory and the receiver matched filter. We use equalizer mean-square error (MSE) as our primary measure of equalizer effectiveness, presented on *heatmap* grids for combinations of cubic vs. linear and quintic vs. cubic adaptive filter memory length. Heatmaps for BPSK, QPSK, 8-PSK, and 16-QAM constellations subjected TWTA nonlinear distortion (and in a representative case, AWGN) are shown. Constellation scatter plots are also presented for each modulation type, using selected values of memory length. Before and after values of SDR and/or signal-to-noise-and-distortion ratio (SNDR) are shown on the constellation plots. Comparison of MSE allows choice of optimum (or near-optimum) memory length values as a practical means of reducing computational burden.

II. LITERATURE REVIEW

TWTAs are commonly used as final amplifiers in satellite microwave communications systems; indeed, a TWTA was used on the first communications satellite, *Telstar I*. Jarrett provides a good introduction to TWTAs in [15]. Continuing development of TWTAs remains an active area of research, with miniaturized TWTAs being developed with cold cathodes and advanced features; see [16], [17], and [18]. For an overview of TWTA models, see Liebtreu [19]. The most widely used TWTA model is the memoryless polynomial Saleh model [20], [21]; modified versions of the Saleh model include [22]. TWTA nonlinear distortion and the Saleh model are frequently used as ''off-the-shelf'' sources of nonlinear distortion for testing various equalization techniques; the sources of various Saleh parameter sets are shown in Table 1.

For a general reference on computational complexity and algorithm analysis, see [23]; see also Floriani's excellent series, *Case Studies in Computer Science* for information on algorithm optimization [24], [25], [26], [27]. Tsimbinos and Lever derive and compare the ''computational complexity'' (see Sections I-D and VI) of the Volterra p^{th}-order inverse (see below) and an adaptive Volterra equalizer [13]. Seagraves et al. find the computational burden of a modified Volterra equalizer (a multilinear-extended-SVD simple Wiener-Volterra equalizer) [12]. Hedge et al. find the computational burden of a nonlinear acoustic channel model with memory and distortion modeled by a Volterra system [28]. Griffith and Arce compute the computational complexity of their RLS partially-decoupled adaptive algorithms (see below), and noted that Volterra kernels can be simplified by symmetry-restriction [7]. Reed and Hawksford [29], like Mathews and Sicuranza [30], noted that Volterra kernels can be simplified by symmetry-restriction.

An essential reference on the Volterra filter is Schetzen's *The Volterra and Wiener Theories of Nonlinear Systems* [31]. See also Schetzen's work on the pth-order inverse [32], which is considered the standard method used to model (adaptively or by statistical characterization) and invert nonlinear systems. This method of inversion was extended by Cheng and Powers, who derived closed-form expressions for the Volterra third- and fifth-order kernel coefficients used in QAM and QPSK nonlinear TWTA channel models [33], [34], [35], [36], [37]. In algebraically deriving their closed-form expressions for kernel coefficients, they used a form of symmetry-restriction to simplify the algebra. Kibangou and Favier found more general expressions using similar methods [38].

For a general reference on communications systems simulation, including on simulating memoryless nonlinearities, see Tranter et al., *Communications Systems Simulation* [39] and Benedetto and Biglieri's *Principles of Digital Transmission* [40]. See Bottomley [41] for an overview of channel equalization. An essential reference on Volterra system modeling is Ogunfunmi's *Adaptive Nonlinear System Identification: the Volterra and Wiener Model Approaches* [42], and his related paper on equalization [43]. Biglieri et al. carried out early work on Volterra equalization of nonlinear channels with TWTA distortion [8], [44], and later derived an optimized the receiver matched filter for nonlinear channel [45]. Effects of *linear* receiver matched filter mismatch were studied by Wickert and Peckham [46]. Haykin's *Adaptive Filter Theory* is the essential reference text on LMS and RLS adaptive filtering [47]. Mathew's *Polynomial Signal Processing* is an essential reference for Volterra filtering (although emphasis is mainly on real-valued applications with quadratic and low-order kernels) [30]. Partial decoupling in Volterra filters is explained by Mathews in [48] and Arenas-García et al. [49]. Griffith developed and tested partially-decoupled adaptive Volterra equalizers for PAM modulation, using even- and odd-order kernels, with TWTA nonlinear AM-AM distortion only (i.e., without AM-PM distortion) [3]. Griffith and Arce developed efficient algorithms for partially-decoupled LMS and RLS equalizers [2], [7]. More recently, Chen et al. studied the effects of a partially-decoupled adaptive equalizer on 16-QAM [4]. Gutierrez and Ryan investigated adaptive (fully-coupled) Volterra equalizers on QPSK, 8-PSK, and QAM channels with TWTA nonlinear distortion, comparing performance to that of a linear adaptive equalizer and noting only slight improvement in BER; importantly, Gutierrez noted that there is no clear relationship between Volterra kernel memory length and equalizer performance [11], [50], [51]. Previous work by the author and advisor used symmetry-restricted Volterra kernels with QPSK modulation in a nonlinear channel with TWTA distortion and AWGN [1]. The present work is an extension thereof.

The authors refer the interested reader to [52] for a more extensive review of the literature.

III. THE TWTA AND NONLINEAR DISTORTION

Traveling-wave tube amplifiers (TWTAs) are one of the few vacuum tube devices still in widespread use. They are highly efficient, high bandwidth microwave amplifiers, offer high gain for little mass, and are largely immune to single-event upsets (SEUs) common in the space environment; they are widely used in satellites as high power RF final amplifiers. When operated at peak efficiency, however, they produce severe nonlinear intermodulation distortion. This nonlinear distortion is described by the *Saleh model*.

A. THE SALEH MODEL

TWTA nonlinear distortion is characterized by the widely-used Saleh model (1), (2), two equations giving the AM-AM and AM-PM intermodulation distortion of a TWTA, respectively, where r is TWTA input power (or in some cases, input *voltage*; e.g, [50]). $A(r)$ is the nonlinear amplitude response, and $\Phi(r)$ is the nonlinear phase response of the TWTA. The Saleh model is a *memoryless* polynomial model.

$$A(r) = \frac{\alpha_A r}{1 + \beta_A r^2} \quad \text{(AM-AM)} \tag{1}$$

$$\Phi(r) = \frac{\alpha_\Phi r^2}{1 + \beta_\Phi r^2} \quad \text{(AM-PM)} \tag{2}$$

When sinusoidal carrier input signal $x(t)$ with angular frequency ω and amplitude- and phase-altering functions $r(t)$ and $\Psi(t)$, is fed into a TWTA, the Saleh functions $A(r)$ and $\Phi(r)$ change the amplitude and phase of the input to produce output signal $y(t)$, as shown in (3) and (4).

$$x(t) = r(t)\cos(\omega t + \Psi(t)) \tag{3}$$

$$y(t) = A(r(t))\cos(\omega t + \Psi(t) + \Phi(r(t))) \tag{4}$$

B. SALEH MODEL PARAMETERS

The amplitude parameters α_A and β_A in (1), and the phase parameters α_Φ and β_Φ in (2) are typically determined empirically by curve-fitting of measured data; however, rounded or theoretical values also may be used. Values of these parameters found in the literature are shown in Table 1. The AM-AM and AM-PM response curves using the *Saleh I* parameter values are plotted in Figure 1. It can be seen in Figure 1 (and Figure 2) that as input power increases, the AM-AM transfer function falls to zero, while the AM-PM function approaches a constant. This is shown in (5) and (6).

TABLE 1. Values of Saleh parameters in the literature.

Name	α_A	β_A	α_Φ	β_Φ	Source
Saleh I	1.9638	0.9945	2.5293	2.8168	[20]
Chen	1.968	0.9945	2.5293	2.8168	[4]
Saleh II	1.6623	0.0553	0.1533	0.3456	[20]
Saleh III	2.1587	1.1517	4.0033	9.104	[20]
Particular	2.1587	1.1517	4.033	9.104	[43]
Tranter	2.1587	1.15	4	2.1	[39]
Typical	2.15	1.15	4	2.1	[19]
Simplified I	2	1	2	1	[19]
Simplified II	2	1	$\pi/3$	1	[34], [53]
Griffith	4	4	0	0	[2]
Karam	2	1	$\pi/6$	1	[54]

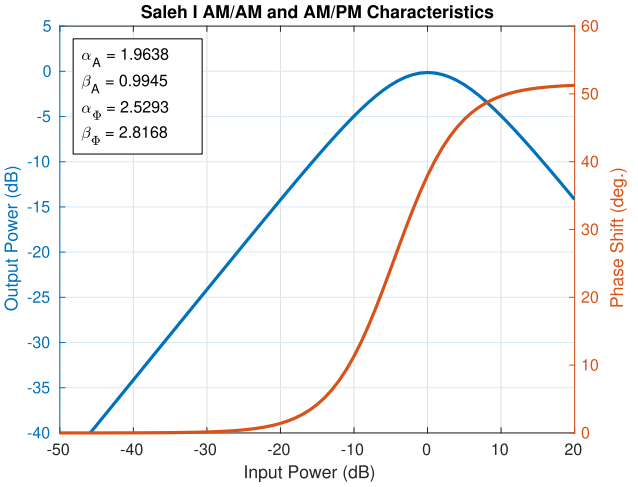


FIGURE 1. Typical TWTA intermodulation distortion characteristics (dB).

$$\lim_{r \to \infty} A(r) = 0, \tag{5}$$

$$\lim_{r \to \infty} \Phi(r) = \frac{\alpha_\Phi}{\beta_\Phi}. \tag{6}$$

The maximum of the Saleh AM-AM function occurs at r_{peak} where power efficiency and AM distortion are maximized, and may be found by setting the derivative of $A(r)$ equal to zero and solving for r, yielding (7):

$$r_{\text{peak}} = \frac{1}{\sqrt{\beta_A}}. \tag{7}$$

This implies that β_A should be unity in the normalized Saleh model; however, deviations from unity are observed in β_A in Tables 1 and 2. The Saleh AM-AM and AM-PM responses of several TWTAs are plotted in Figure 2. It can be seen that the TWTAs fall into three general groups according to their limiting phase shift; these groups are ∼30°, ∼60°, and ∼110°. We note that TWTA nonlinear phase distortion is particularly troublesome; while some variation in amplitude can be tolerated in PSK signals, equalization of nonlinear variation in phase is more challenging.

Figure 3 illustrates the effect of TWTA distortion. Values on the identity lines $Q = \pm I$ of the complex plane experience an amplitude-dependent counterclockwise rotation, causing the pinwheel pattern in the figure. TWTA nonlinear distortion has the effect of *rotating* a PSK constellation, while *warping* a QAM constellation [40]. (Griffith and Arce used an unnormalized Saleh model with no phase distortion because they were equalizing a PAM system [2], [3], [7].)

C. THE LNL FILTER CASCADE

In simulating communications systems, the Saleh nonlinearity is normally sandwiched between two low-pass linear filters. This is illustrated in Figure 4. The low-pass filters simulate analog "zonal" filters in the communications system and are used to reduce spectral regrowth caused by the nonlinearity, suppressing harmonics and spurious

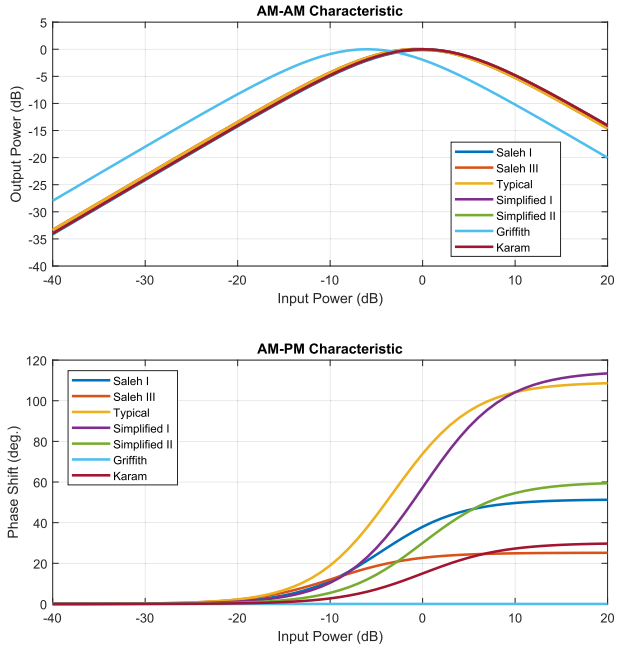


FIGURE 2. Saleh AM-AM and AM-PM responses (in dB).

TABLE 2. Peak input and maximum phase shifts of the Saleh models.

Name	r_{peak}	Φ_{max}
Saleh I	1.00276	51.45°
Chen	1.00276	51.45°
Saleh II	4.25243	25.40°
Saleh III	0.93181	25.20°
Particular	0.93181	25.38°
Tranter	0.93250	109.13°
Typical	0.93250	109.13°
Simplified I	1	114.59°
Simplified II	1	60°
Griffith	0.5	0°
Karam	1	30°

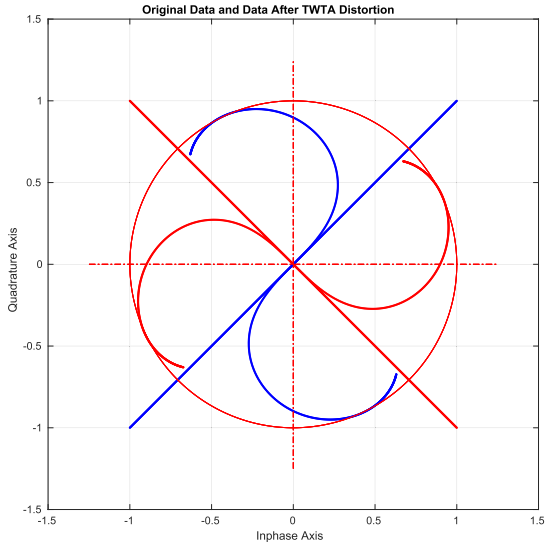


FIGURE 3. "Pinwheel effect" caused by TWTA distortion.

emissions. Butterworth or Chebyshev filters are typically used in simulations; we used Butterworth filters exclusively. These filters add memory (and frequency selectivity) to

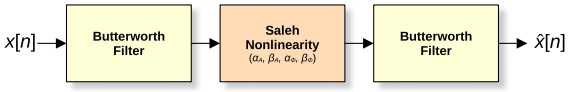


FIGURE 4. The LNL filter cascade used to generate simulation data.

the nonlinear channel. The order and cutoff frequency of these filters found in the literature varies widely; some typical values of pre- and post-filter parameters found in the literature are shown in Table 3. The orders and cutoff frequencies of the filters used in the simulations should be chosen to match closely the analog filters used in the satellite hardware. As will be seen, the order of the pre- and

TABLE 3. Values of pre- and post-filter parameters found in the literature.

Source	Pre-Filter		Post-Filter	
	Order	−3 dB Cutoff	Order	−3 dB Cutoff
[8], [40]	6	$1.10/T_s$	4	$1.20/T_s$
[8], [40]	6	$1.10/T_s$	2	$1.20/T_s$
[11], [50]	6	$0.75/T_s$	6	$0.75/T_s$
[1]	5	$0.70/T_s$	7	$0.80/T_s$

post-filters largely control the variance and thus the SDR of the distorted constellation. If the variance of the distorted constellation is sufficiently high and non-homogeneous in the *circumferential direction* along the unit circle of the I-Q plane (with or without AWGN), the equalized constellation will form a crescent-like or multilobate pattern with little improvement in BER; however, if the pattern is relatively homogeneous, the equalized constellation will form closely-spaced circular or elliptical patterns. It will also be shown that the receiver matched filter greatly exaggerates these tendencies.

IV. THE VOLTERRA FILTER

Equation (8) is the most general form of a fifth-order (*quintic*) truncated FIR Volterra filter. The authors call this form the "canonical" Volterra filter, as it is this form that invariably appears in the literature. Square brackets are used to indicate that respective filter input and output sequences $x[n]$ and $y[n]$ are discrete (and complex-valued). Array h_1 and multi-dimensional arrays h_3, and h_5 are the respective linear, cubic, and quintic kernels, each holding complex-valued filter coefficients. The Volterra filter is nonlinear in that its output contains *products* of input sequence terms as well as the familiar linear combinations of input terms generated by the linear kernel. This filter is a natural choice for equalizing or modeling TWTA nonlinear distortion (and for modeling many other nonlinear phenomena).

$$y[n] = \sum_{i=0}^{N_1-1} h_1(i)\, x[n-i] + \sum_{i=0}^{N_3-1}\sum_{j=0}^{N_3-1}\sum_{k=0}^{N_3-1} h_3(i,j,k)\, x^*[n-i]\, x[n-j]\, x[n-k]$$

$$+\sum_{i=0}^{N_5-1}\sum_{j=0}^{N_5-1}\sum_{k=0}^{N_5-1}\sum_{l=0}^{N_5-1}\sum_{m=0}^{N_5-1} h_5(i,j,k,l,m)\,x^*[n-i]\,x^*[n-j]\,x[n-k]\,x[n-l]\,x[n-m] \quad (8)$$

It is known that only the linear, third-order (*cubic*) and fifth-order (*quintic*) convolutions are needed to equalize (or model) TWTA distortion, as even-order Volterra kernels produce out-of-band signals [30], [42]. Further, Volterra equalization of constant-power modulation schemes such as M-PSK require *only* the linear and cubic kernels for equalization; the quintic kernel may be excluded for greater computational efficiency. While very good equalization of M-QAM signals can be achieved with linear and cubic kernels, all three kernels—linear, cubic, and quintic—are necessary for *optimal* equalization of M-QAM signals. (Our simulation experiments show that including the quintic kernel when equalizing 16-QAM typically reduces MSE by ∼0.5 dB over that provided by a cubic Volterra equalizer.)

A. MEMORY LENGTH

Parameters N_1, N_3, and N_5 in (8) are the respective linear, cubic, and quintic *memory lengths* of each kernel (hereinafter abbreviated as $N = [N_1, N_3, N_5]$ for convenience). In the canonical form (8), each index variable runs from zero to one less than memory length. Choice of cubic and quintic memory length is critical for minimizing computational burden because the number of filter coefficients in the cubic and quintic kernels grows geometrically with cubic and quintic memory length (as N_3^3 and N_5^5, respectively), and quickly becomes unmanageable even for modest values of N_3 and N_5. We therefore seek to curb the geometric growth in number of kernel coefficients by eliminating *unnecessary* coefficients from the cubic and quintic kernels. We further desire to *minimize* the memory lengths used in all orders of the equalizer by empirically choosing the smallest number of memory units needed for a given degree of equalization (see Section VIII).

B. SYMMETRY-RESTRICTION

The canonical Volterra filter (8) is *overparameterized* in that the cubic and quintic kernels h_3 and h_5 contain many *redundant* coefficients, a result of the commutative and associative properties of multiplication in the factors of the cubic and quintic convolution summands, yielding *symmetric* kernels with coefficients multiplying each *permutation* of summand factors. Efficient parametrization of the Volterra filter may be viewed as a combinatorics problem [30]. Redundant coefficients are removed by altering the summation limits as in (9), yielding *symmetry-restricted* cubic and quintic kernels with far fewer cubic and quintic kernel coefficients than are found in the canonical form.

$$y[n] = \sum_{i=0}^{N_1-1} h_1(i)\,x[n-i]$$
$$+\sum_{i=0}^{N_3-1}\sum_{j=0}^{N_3-1}\sum_{k=0}^{j} h_3(i,j,k)\,x^*[n-i]\,x[n-j]\,x[n-k]$$
$$+\sum_{i=0}^{N_5-1}\sum_{j=0}^{i}\sum_{k=0}^{N_5-1}\sum_{l=0}^{k}\sum_{m=0}^{l} h_5(i,j,k,l,m)\,x^*[n-i]\,x^*[n-j]\,x[n-k]\,x[n-l]\,x[n-m] \quad (9)$$

The symmetry-restricted Volterra kernel coefficients in (9) each correspond to one *combination* of summand factors. Elimination of redundant coefficients by symmetry-restriction greatly increases computational efficiency without affecting filter action.

C. CONVOLUTION SUMMAND FACTORING

Inspection of the cubic and quintic convolutions in both the canonical form (8) and the symmetry-restricted form (9) of the Volterra filter reveals that their summands may be factored to associate dependent input values with their respective summations. This allows for significant gains in computational efficiency by eliminating unnecessary multiplications and additions. The symmetry-restricted, factored form is (10).

$$y[n] = \sum_{i=0}^{N_1-1} h_1(i)\,x[n-i]$$
$$+\sum_{i=0}^{N_3-1} x^*[n-i]\sum_{j=0}^{N_3-1} x[n-j]\sum_{k=0}^{j} h_3(i,j,k)\,x[n-k]$$
$$+\sum_{i=0}^{N_5-1} x^*[n-i]\sum_{j=0}^{i} x^*[n-j]\sum_{k=0}^{N_5-1} x[n-k]\sum_{l=0}^{k} x[n-l]\sum_{m=0}^{l} h_5(i,j,k,l,m)\,x[n-m] \quad (10)$$

This is the form used in our simulations, as it is far more computationally efficient than the canonical form (8), the canonical factored form, or the (non-factored) symmetry-restricted form (9). The computational efficiency of these forms are compared in Section VI.

V. THE LMS ADAPTIVE VOLTERRA EQUALIZER

In this section we develop the odd-order quintic nonlinear LMS adaptive Volterra equalizer used in our simulations. To simplify notation, in this section we use $x[n]$ and $y[n]$ to denote the respective (complex-valued) adaptive equalizer input and output sequences. (In Section VII we will use $z[n]$ to denote the distorted and possibly noisy recovered symbol sequence used as equalizer input, while equalizer output is denoted by $y[n]$; see Figure 8.)

The *desired* sequence $d[n]$ is used for training the adaptive kernel coefficients, and is known *a priori* (e.g., a known training sequence is transmitted from the satellite, which, in practice, may comprise up to 20% of transmitted data [55]). The error sequence $e[n]$ is the difference between the desired

sequence and the filter output:

$$e[n] = d[n] - y[n]. \tag{11}$$

The LMS adaptive algorithm minimizes the mean-square-error (MSE) between the desired and actual equalizer output:

$$\text{MSE} = E\left[e^*[n]\, e[n]\right] = E\left[\left(d[n] - y[n]\right)^2\right]. \tag{12}$$

A. THE FULLY-COUPLED EQUALIZER

The standard (linear) LMS algorithm is altered to accommodate the nonlinear Volterra kernels h_3 and h_5 by including separate state vectors $\mathbf{x}_1[n]$, $\mathbf{x}_3[n]$, and $\mathbf{x}_5[n]$ as in (13), (14), and (15). The state vectors of each order consist of terms, and products of terms, in the input sequence $x[n]$.

$$\begin{aligned}\mathbf{x}_1[n] &= [x[n], x[n-1], x[n-2], x[n-3], \ldots, \\ &\quad x[n-N_1-1]\,]^\mathsf{T} \end{aligned} \tag{13}$$

$$\begin{aligned}\mathbf{x}_3[n] &= \big[x^*[n]x^2[n], x^*[n]x[n]x[n-1], \ldots, \\ &\quad x^*[n-N_3-1]x^2[n-N_3-1]\,\big]^\mathsf{T} \end{aligned} \tag{14}$$

$$\begin{aligned}\mathbf{x}_5[n] &= \big[(x^*[n])^2x^3[n], x^*[n-1]x^*[n]x^3[n], \ldots, \\ &\quad (x^*[n-N_5-1])^2x^3[n-N_5-1]\,\big]^\mathsf{T} \end{aligned} \tag{15}$$

Equalizer output $y[n]$ is computed as (16). Each kernel is stored as a column vector $\mathbf{h}_p$ where $p = 1, 3, 5$.

$$y[n] = \mathbf{h}_1^\mathsf{T}[n]\mathbf{x}_1[n] + \mathbf{h}_3^\mathsf{T}[n]\mathbf{x}_3[n] + \mathbf{h}_5^\mathsf{T}[n]\mathbf{x}_5[n] \tag{16}$$

Kernel arrays $\mathbf{h}_1$, $\mathbf{h}_3$, and $\mathbf{h}_5$ are each updated separately, as per (17), (18), and (19):

$$\mathbf{h}_1[n+1] = \mathbf{h}_1[n] + 2\mu_1 e[n]\mathbf{x}_1^*[n], \tag{17}$$

$$\mathbf{h}_3[n+1] = \mathbf{h}_3[n] + 2\mu_3 e[n]\mathbf{x}_3^*[n], \text{ and} \tag{18}$$

$$\mathbf{h}_5[n+1] = \mathbf{h}_5[n] + 2\mu_5 e[n]\mathbf{x}_5^*[n], \tag{19}$$

where μ_1, μ_3, and μ_3 are the step sizes used on each order. Fixed LMS step sizes of $\mu_1 = 10^{-3}$, $\mu_3 = 10^{-4}$, and $\mu_5 = 10^{-5}$, were used in all simulations. Extensive experimentation showed that these values yielded good results; they match the step sizes used by Gutierrez and Ryan [11]. In this configuration, the LMS Volterra equalizer operates as a *linear-in-the-parameters* or *pseudolinear* filter, wherein the filter is linear with respect to the output sequence $y[n]$.

The state vectors and kernels can be concatenated to form a single input vector $\mathbf{x}$ and a single kernel vector $\mathbf{h}$ as in (20) and (21), respectively, such that a single vector multiplication forms the final convolution product (22). A single equation (23) is then used to perform the kernel update.

$$\mathbf{x}[n] = \left[\mathbf{x}_1^\mathsf{T}, \mathbf{x}_3^\mathsf{T}, \mathbf{x}_5^\mathsf{T}\right]^\mathsf{T} \tag{20}$$

$$\mathbf{h}[n] = \left[\mathbf{h}_1^\mathsf{T}, \mathbf{h}_3^\mathsf{T}, \mathbf{h}_5^\mathsf{T}\right]^\mathsf{T} \tag{21}$$

$$y[n] = \mathbf{h}^\mathsf{T}[n]\,\mathbf{x}[n] \tag{22}$$

$$\mathbf{h}[n+1] = \mathbf{h}[n] + 2\mu e^*[n]\,\mathbf{x}[n] \tag{23}$$

If the filter is implemented in this manner, the step-size matrix μ in (23) holds scalar step-sizes μ_1, μ_3, and μ_5 in the main diagonal of an (otherwise zero) square matrix. In practice, however, μ is structured as a column vector, and the second term of (23) is implemented using element-by-element multiplication with $\mathbf{x}[n]$ for increased computational efficiency. However, as will be seen in Section V-B, segregating the kernel and state vectors by order facilitates partial-decoupling of the kernels.

Note that in (17), (18), and (19), the *state vectors* are conjugated. This differs from the LMS algorithm commonly found in textbooks, in which the *error* is conjugated and the state vector(s) are not. *Either* the state vectors *or* the error must be conjugated to maintain the real-imaginary cross-coupling needed by the LMS algorithm [47, p. 239]. If the state vectors are conjugated, (16) holds and the *transpose* of each kernel is multiplied by its respective state vector. If the error is conjugated, the Hermitian is used instead of the transpose, as in (24). The linear form of (24) is the usual form found in textbooks (e.g., [47]).

$$y[n] = \mathbf{h}_1^\mathsf{H}[n]\mathbf{x}_1[n] + \mathbf{h}_3^\mathsf{H}[n]\mathbf{x}_3[n] + \mathbf{h}_5^\mathsf{H}[n]\mathbf{x}_5[n]. \tag{24}$$

While either form may be used, the respective kernels generated by each form will be complex-conjugates of each other. It must be remembered that the conventional "linear" LMS adaptive algorithm (so-called because linear convolution is used to compute the filter output) is itself a *nonlinear* filter which violates the principles of homogeneity and superposition [47, p. 259]. By modifying the LMS algorithm to include the cubic- and quintic-order kernels, additional degrees of nonlinearity are introduced, making the LMS adaptive Volterra equalizer used herein a *highly* nonlinear filter.

In a conventional adaptive nonlinear Volterra filter, such as (16), where the linear, cubic, and quintic kernels are updated using (17), (18), and (19), respectively, the kernels of each order influence each other during convergence; the kernels are said to be *fully-coupled*. This is illustrated in Figure 5.

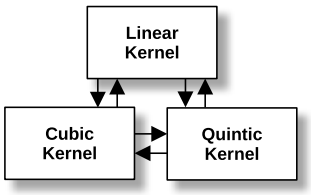


FIGURE 5. Fully-coupled kernels influence each other during convergence.

B. THE PARTIALLY-DECOUPLED EQUALIZER

Volterra kernels may be *partially-decoupled* by altering the sequence in which the error is calculated and the kernel updates are performed. In a partially-decoupled LMS adaptive filter, the kernels of each order are applied *sequentially*. The error is calculated and coefficients are updated for each respective order. This is shown in (25), (26),

and (27) for the linear case:

$$y[n] = \mathbf{h}_1^{\mathsf{T}}[n]\mathbf{x}_1[n]\,, \tag{25}$$
$$e[n] = d[n] \; - \; y[n], \tag{26}$$
$$\mathbf{h}_1\,[n+1] = \mathbf{h}_1[n] \; + \; 2\mu_1 e[n]\mathbf{x}_1^*\,[n]. \tag{27}$$

The cubic contribution is then added to the output variable $y[n]$, the error recalculated, and the cubic kernel updated, as in (28), (29), and (30):

$$y[n] \Leftarrow y[n] \; + \; \mathbf{h}_3^{\mathsf{T}}[n]\mathbf{x}_3[n]\,, \tag{28}$$
$$e[n] = d[n] \; - \; y[n]\,, \tag{29}$$
$$\mathbf{h}_3\,[n+1] = \mathbf{h}_3[n] \; + \; 2\mu_3 e[n]\mathbf{x}_3^*\,[n]\,. \tag{30}$$

Next, the quintic contribution is added to the output, the error recalculated, and the quintic kernel updated *in simili modo*:

$$y[n] \Leftarrow y[n] \; + \; \mathbf{h}_5^{\mathsf{T}}[n]\mathbf{x}_5[n]\,, \tag{31}$$
$$e[n] = d[n] \; - \; y[n], \tag{32}$$
$$\mathbf{h}_5\,[n+1] = \mathbf{h}_5\,[n] \; + \; 2\mu_5 e[n]\mathbf{x}_5^*\,[n]. \tag{33}$$

Calculating the error and updating the kernels in this manner forces unidirectional kernel coupling, illustrated in Figure 6; during convergence, lower-order kernels influence the higher-order kernels, but not vice versa.

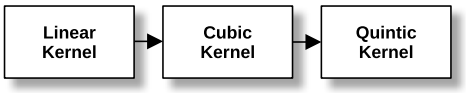


FIGURE 6. Partially-decoupled kernels exhibit unidirectional influence.

Partial-decoupling offers several advantages: 1) kernels of different order can be adjusted quasi-independently, 2) kernels of all orders are significantly less prone to the effects of outliers, 3) higher-order kernels can be included or excluded as needed, 4) maximum LMS step size can be computed exactly for each order of kernel, and 5) partially-decoupled kernels tend to converge faster than fully-coupled kernels. The unidirectional kernel update process is more intuitive and easier to grasp than the complex kernel interaction which takes place during convergence of a fully-coupled adaptive Volterra filter. These advantages, however, come at the cost of optimization: partially-decoupled adaptive Volterra filters are not optimized with respect to MSE and will therefore converge to a slightly higher MSE than their fully-coupled counterparts.

In contrast to fully-coupled adaptive Volterra filters, partially-decoupled adaptive Volterra equalizers have received little attention in the literature. Partial decoupling of Volterra kernels was first described by Mathews [48]. Griffith and Arce introduced partially-decoupled LMS and RLS adaptive Volterra equalizers and tested them with real-valued M-PAM signals with TWTA AM-AM distortion [2], [3], [7].

As will be seen in Section VIII, our simulation results indicate that for M-PSK modulation, coupling mode mainly affects equalization in the *radial* direction of the I-Q plane, with fully-coupled and partially-decoupled LMS equalizers converging to relatively similar MSE values. However, for 16-QAM modulation, fully-coupled equalizers performed significantly better than their partially-decoupled counterparts.

TABLE 4. Kernel coefficients in canonical and symmetry-restricted forms.

Form	Lin.	Cubic	Quintic
Can.	N_1	N_3^3	N_5^5
S-R	N_1	$\left(N_3^3 + N_3^2\right)/2$	$\left(N_5^5 + 4N_5^4 + 5N_5^3 + 2N_5^2\right)/12$

TABLE 5. Reduction in computational burden for $N = [5, 5, 5]$.

Form	Multiplies	Adds	$-\Delta$ Mult.	$-\Delta$ Adds
Canonical	64,020	38,520	-	-
Can. Fact.	16,260	16,260	74.6%	57.8%
S-R	11,420	6,920	82.2%	82.0%
S-R Fact.	3,820	3,820	94.0%	90.1%

VI. COMPUTATIONAL EFFICIENCY

In Section IV we introduced two forms the Volterra filter, the canonical form (8) and the symmetry-restricted form (9); each of these forms, when trained adaptively, give equivalent results, with the symmetry-restricted form being more computationally-efficient by virtue of having fewer coefficients. We also noted that each of these forms may be factored for a significant gain in computational efficiency. We now consider the computational efficiency of each of these forms of the Volterra filter by counting the number of kernel coefficients as well as the number of multiplications and additions incurred by each form per iteration of the filter.

Table 4 shows polynomial expressions for the number of complex-valued coefficients in the linear, cubic, and quintic kernels of the canonical and symmetry-restricted forms of the Volterra filter as functions of memory length. The reduced number of kernel coefficients in the symmetry-restricted form yields a significant gain in computational efficiency when compared to the canonical form.

Further gains in computational efficiency may be realized by factoring the cubic and quintic convolutions, as is done in (10). Polynomial expressions for computational burden (i.e., the number of multiplication operations per iteration of the filter) of four forms of the Volterra filter (with and without symmetry-restriction and factoring) are shown in Table 6. These polynomial functions are plotted against typical values of memory length (ranging from 1 to 10, where $N_3 = N_5$) in Figure 7. Quintic (dashed lines) and cubic (solid lines) terms are plotted separately. It can be seen that factoring and symmetry-restriction together significantly reduce computational burden, typically by $\sim$1 order-of-magnitude. While less important than multiplications, polynomial expressions for number of additions per iteration are shown Table 7.

Table 5 shows the total number (linear, cubic, and quintic) of real-valued multiplications and additions and their percentage reductions using typical memory length values of $N_1 = 5$, $N_3 = 5$, and $N_5 = 5$ ($N = [5, 5, 5]$), for each

TABLE 6. Computational burden (multiplication) of various forms of the Volterra filter.

Form	Multiplications per Iteration
Canonical	$4\left[N_1 + 3N_3^3 + 5N_5^5\right]$
Canonical Factored	$4\left[N_1 + \left(N_3^3 + N_3^2 + N_3\right) + \left(N_5^5 + N_5^4 + N_5^3 + N_5^2 + N_5\right)\right]$
Symmetry-Restricted	$4\left[N_1 + \frac{3}{2}\left(N_3^3 + N_3^2\right) + \frac{5}{12}\left(N_5^5 + 4N_5^4 + 5N_5^3 + 2N_5^2\right)\right]$
Symmetry-Restricted Factored	$4\left[N_1 + \left(\frac{1}{2}N_3^3 + \frac{3}{2}N_3^2 + N_3\right) + \frac{1}{12}\left(N_5^5 + 7N_5^4 + 17N_5^3 + 17N_5^2 + 18N_5\right)\right]$

TABLE 7. Computational burden (addition) of various forms of the Volterra filter.

Form	Additions per Iteration
Canonical	$2\left[2N_1 + 4N_3^3 + 6N_5^5\right]$
Canonical Factored	$2\left[2N_1 + 2\left(N_3^3 + N_3^2 + N_3\right) + 2\left(N_5^5 + N_5^4 + N_5^3 + N_5^2 + N_5\right)\right]$
Symmetry-Restricted	$2\left[2N_1 + 2\left(N_3^3 + N_3^2\right) + \frac{1}{2}\left(N_5^5 + 4N_5^4 + 5N_5^3 + 2N_5^2\right)\right]$
Symmetry-Restricted Factored	$2\left[2N_1 + \left(N_3^3 + 3N_3^2 + 2N_3\right) + \frac{1}{6}\left(N_5^5 + 7N_5^4 + 17N_5^3 + 17N_5^2 + 18N_5\right)\right]$

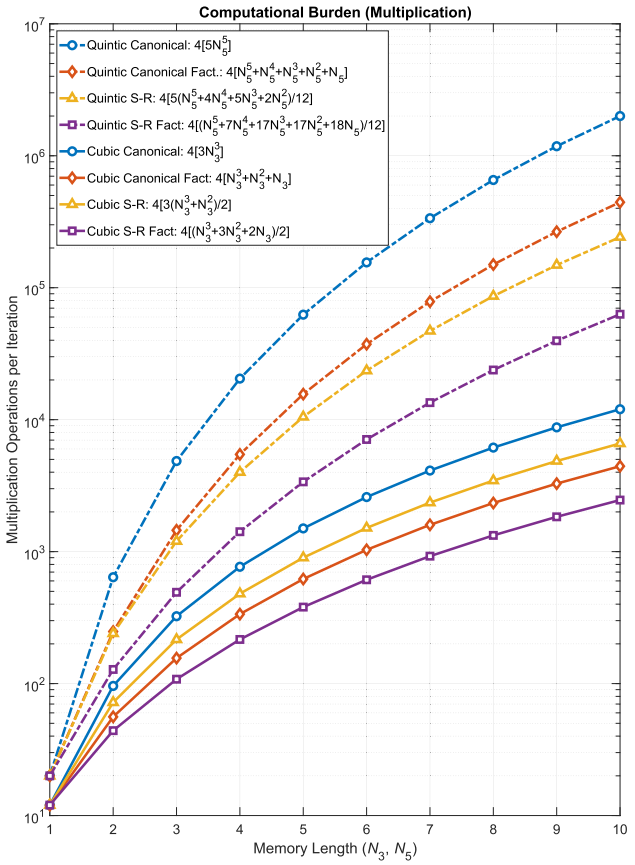


FIGURE 7. Cubic and quintic computational burden vs. memory length.

form of the filter. It can be seen that factoring and symmetry restriction each significantly reduces computational burden; when combined, these strategies result in a substantial ∼94% reduction in multiplicative computational burden, highly desirable for efficient FPGA implementation.

While we use an *adaptive* LMS Volterra equalizer in this research, we note that the techniques of symmetry restriction and factoring may be applied to *any* implementation of the Volterra filter for greatly increased computational efficiency.

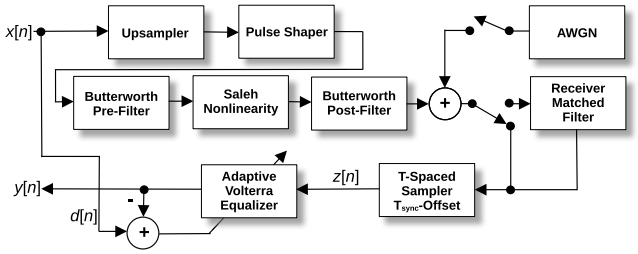


FIGURE 8. Communication system simulation model.

VII. THE SIMULATION MODEL

The simulation experiments used the baseband system model shown in Figure 8. A pseudorandom sequence of complex-valued symbols $x[n]$ produced by a conventional BPSK, QPSK, 8-PSK, or 16-QAM baseband modulator (not shown) is upsampled at $N_s = 10$ samples per symbol, then fed to a pulse shaper to produce a rectangular pulse. Rectangular pulse shapes were used exclusively in these simulations. The signal is then fed to the LNL filter cascade (described previously in Section III-C) consisting of a low-pass Butterworth pre-filter, the *Saleh I* nonlinearity, and a low-pass Butterworth post-filter, representing the nonlinear channel. AWGN may be optionally added to the signal (we consider the effects of AWGN in Section VIII-A). The distorted and filtered signal is then optionally processed by a receiver matched filter before being re-sampled at symbol-spacing with optimal offset T_{sync} to recover the transmitted symbol sequence $z[n]$. Next, the recovered symbol sequence $z[n]$ is processed by the fully-coupled or partially-decoupled LMS adaptive Volterra equalizer to produce the equalized output sequence $y[n]$.

The equalizer is trained using the undistorted input symbol sequence $x[n] = d[n]$, after compensating for phase lag between $x[n]$ and $z[n]$ (determined by the order of the filters in the LNL cascade). The first 80% of the symbol sequence is used for training; the final 20% is used for evaluating equalizer performance and is plotted in the scatter plots in Section VIII. Unless otherwise specified, 10^6 symbols were used in each simulation.

A. SIGNAL-TO-DISTORTION RATIO

We calculate the signal-to-distortion ratio (SDR) from the measured error-vector-magnitude (EVM) of the distorted (unequalized) constellation using the following relationship:

$$\mathrm{SDR_{dB}} = -\left(\mathrm{PAPR} + 20\log_{10}\left(\frac{\mathrm{EVM_{\%}}}{100}\right)\right) \tag{34}$$

where PAPR is the peak-to-average-power-ratio for the constellation under consideration. Table 8 shows the PAPR values for the modulation schemes used herein. Because the simulations are conducted at baseband, we are unconcerned with the PAPR of the transmitted sinusoidal waveforms and need only consider the PAPR of the constellations used in the simulations. For constant-amplitude M-PSK constellations, peak-to-average power ratio is 1:1 (0 dB); only the 16-QAM constellation has a different PAPR. (For a good explanation of how EVM relates to other performance metrics see [56].)

TABLE 8. PAPR values for various modulation schemes.

Modulation Type	PAPR (dB)
BPSK	0
QPSK	0
8-PSK	0
16-QAM	2.55

B. COUNTER-ROTATION

The adaptive Volterra equalizer rotates the distorted constellation points (clockwise) to their desired location in the I-Q plane. However, in practice there are other means of rotating the constellation (e.g., carrier-tracking), and we are more interested in the equalizer's ability to reduce variance about the ideal symbol points in the distorted constellation, especially moving symbols away from the decision boundaries. For purposes of evaluating equalizer performance, we therefore mimic carrier-tracking by *artificially* counter-rotating the distorted M-PSK constellation back to its starting point prior to calculating EVM. This artificial counter-rotation is reflected in the scatter plots for BPSK, QPSK, and 8-PSK shown in Section VIII, wherein the distorted clouds of blue symbols were counter-rotated to their starting positions (e.g., their original quadrants for QPSK) for a realistic comparison with the equalized (red) symbols. However, counter-rotation was *not* used on the unequalized 16-QAM constellations, as the warping effect of the TWTA is amplitude-dependent and much more significant for QAM; compensation for warping cannot be approximated by a simple counter-rotation in a manner analogous to carrier-tracking.

Applying (34) to the calculated EVM of symbols in the distorted, counter-rotated constellation gives SDR in units of dB directly comparable to SNR. Equation (34) is also used with distorted data having AWGN to find the signal-to-noise-and-distortion-ratio (SNDR). It must be remembered that while the distorted symbols appear as a cloud in the scatter plots, unlike AWGN, the distortion is *deterministic*; it is only stochastic in the sense that the transitions between symbols are pseudorandom.

C. CHANNEL MEMORY AND THE RECEIVER MF

To better understand the effects of memory in the nonlinear channel, several combinations of pre- and post-filters of various orders, both with and without a receiver matched filter (MF), were tested. 5×10^5 QPSK symbols with rectangular pulse shape and $N_s = 10$ samples per symbol were processed by the LNL filter cascade using the *Saleh I* parameters at full saturation. No AWGN was used in these studies. RMS EVM was calculated after artificial counter-rotation for each distorted constellation, and (34) was used to convert EVM to SDR. Combinations of filter order were selected (ranging from first- to ninth-order), covering all values found in the literature, including those used by the author and advisor in previous work [1]. In all cases, Butterworth low-pass filters with normalized bandwidth cutoffs of $0.75/T_s$ were used. Results are summarized in Table 9. The SDR values are comparable to SNR, if AWGN rather than distortion had been imposed on the constellation. This gives the reader an intuitive idea of the magnitude of distortion to be equalized.

TABLE 9. EMV and SDR for various orders of LNL pre- and post-filters.

Order		With Receiver MF		Without Receiver MF	
Pre	Post	EVM (%)	SDR (dB)	EVM (%)	SDR (dB)
1	1	18.97	14.44	4.61	26.72
1	2	15.34	16.29	3.86	28.27
2	2	14.09	17.02	5.21	25.66
2	3	13.74	17.24	8.00	21.93
3	3	14.43	16.81	10.03	19.97
3	4	14.92	16.52	12.15	18.31
4	4	15.81	16.02	13.43	17.44
4	5	16.51	15.65	15.08	16.43
5	5	17.07	15.35	16.09	15.87
5	6	18.16	14.82	17.46	15.16
5	7	18.92	14.46	19.96	14.44
6	6	18.36	14.72	18.54	14.64
6	7	19.45	14.22	19.80	14.07
7	7	20.01	13.98	20.56	13.74
7	8	20.77	13.65	21.98	13.16
7	9	21.88	13.20	23.15	12.71

It is evident from the data in Table 9 that all combinations of filter orders examined introduce significant distortion (∼15 dB SDR) when *used with a matched filter*. Without the matched filter, filters of order ∼5 or more introduce distortion close to this level. Lower-order filters introduce significantly less distortion, explaining why disparate results are often reported in the literature (e.g., [9] compared to [1]). However, *linear* adaptive equalization is often adequate for channels with SDR (or SNR) >20 dB for QPSK, where the symbols are widely separated; the closer symbol spacing in 8-PSK and QAM requires greater equalization.

Including a receiver matched filter is standard practice in simulating communication systems, as it provides an optimal match to the transmitted waveform. However, the nonlinear

channel significantly distorts the original waveform, making the match less than optimal. In testing Volterra equalizers, Gutierrez and Ryan [11] noted this problem and used a fractionally-spaced LMS adaptive *linear* pre-filter prior to their LMS adaptive Volterra equalizer, and noted a slight improvement in BER compared to a conventional receiver MF. Biglieri et al. derived the *optimal* receiver matched filter for a nonlinear channel [45].

Instead of using these techniques, we omitted the receiver MF entirely, hypothesizing that the linear kernel of the adaptive Volterra equalizer, if provided with sufficient memory, would converge so as to match the distorted waveform to the channel, particularly in the partially-decoupled case. Results reported in Section VIII support this hypothesis. While the match is likely not optimal, it is better than that obtained with a conventional receiver MF, which, as will be shown, exaggerates *circumferential* variance in the distorted constellation, whereas the adaptive Volterra filter is most effective in the *radial* direction. It is for this reason that a lower MSE does not necessarily translate to a lower BER. *Quidquid recipitur secundum modum recipientis recipitur.*

VIII. SIMULATION RESULTS

In this section we present our simulation results for BPSK, QPSK, 8-PSK, and 16-QAM, using the system model presented in Figure 8 of Section VII. Fifth- and seventh-order Butterworth low pass filters, each having normalized cutoff frequencies of $0.75/T_s$ were used as the LNL pre- and post-filters, respectively, with the *Saleh I* parameter set used in the Saleh model. These values were selected to produce as much distortion as the equalizer could realistically tolerate, as adding memory of order six or more tended to produce a distortion cloud difficult to equalize. For select cases of QPSK, results from both fully-coupled and partially-decoupled equalizers are shown to illustrate the effect of kernel coupling. In the PSK simulations, 0 dB input backoff was used, putting the Saleh model in full saturation; in the QAM simulations, a realistic value of 6 dB IBO was used, as some degree of IBO must be used for transmitting QAM with a TWTA. For purposes of comparison, the receiver matched filter was excluded from all simulations except those presented below in Section VIII-A. Fixed LMS step sizes of $\mu_1 = 10^{-3}$, $\mu_3 = 10^{-4}$, and $\mu_5 = 10^{-5}$ were used in all simulations; 10^6 symbols were used in each simulation.

A. THE EFFECTS OF AWGN AND CHANNEL MEMORY

Using QPSK as an example, we first consider how including AWGN affects the equalized constellation.

Previous work by the author and advisor [1] demonstrated that an LMS adaptive Volterra equalizer was only marginally effective at reducing SEP in QPSK with AWGN. Figure 9 shows the action of an LMS adaptive Volterra equalizer with TWTA nonlinear distortion and AWGN at an SNR of 15 dB, giving an SNDR of 11.67 dB. Memory lengths $N = [3, 3, 0]$ were used in this equalizer, with a conventional receiver MF. The prominent *crescent patterns* in the equalized constellation reveal much about equalizer behavior. The limbs of the crescents fall slightly *outside* of the unit circle, while the tips of the crescents are relatively unaffected by the equalizer and remain close to the decision boundaries. It is for this reason that the equalizer was found to be of little use in reducing SEP. Gutierrez and Ryan observed that when the SNR level exceeds the SDR, equalization takes place *in the radial direction only*, which does not translate into reduced BER [11]. We have observed that the equalizer tends to form crescent patterns at SNR levels of ∼20 dB or lower, but not at SNR of ∼30 dB or higher. Increasing noise energy tends to blur the equalized constellation into crescents.

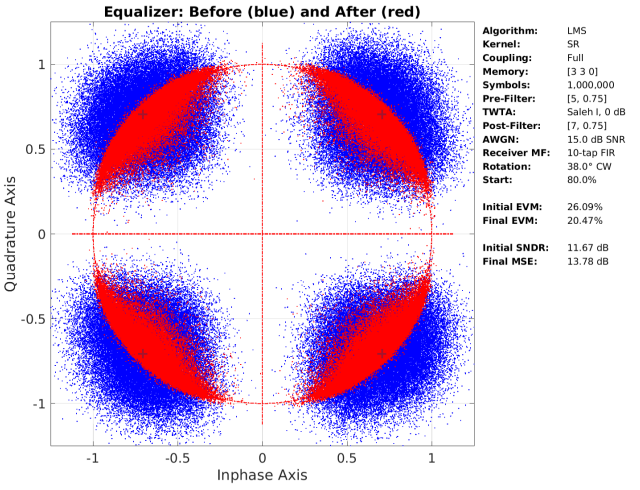


FIGURE 9. Crescent patterns in equalized QPSK constellation, 15 dB SNR.

Next we consider the same unequalized constellation with a lower noise energy. Figure 10 shows results of an otherwise identical simulation but using 30 dB SNR, a level well below the SDR. We see that the crescents have been replaced by *trilobate* patterns which extend farther outside the unit circle than the crescents. Removing *all* noise results in the pattern in Figure 11; the trilobate patterns have been replaced by individual symbol clouds on the unit circle. Again, we see that equalization occurs mainly in the radial direction.

Figure 12 shows results of the same simulation but using only 10^5 symbols. With an order-of-magnitude fewer symbols used to train the equalizer, equalization in the radial direction occurs more slowly than in the previous case. Further, radial equalization *inside* the unit circle is significantly slower than that outside of the unit circle. We refer to this phenomenon as the "slow radial convergence mode." Long training sequences (∼10^6 or more symbols) are needed to reveal this property of the equalizer; it may easily be overlooked, as training sequences reported in the literature typically consist of ∼10^4 symbols.

From these examples we see that the distribution of symbols in the distorted constellation significantly influences equalizer behavior. Aside from the constellation rotation (and warping) caused by the Saleh TWTA model, it is the *memory in the nonlinear channel,* i.e., the order of the

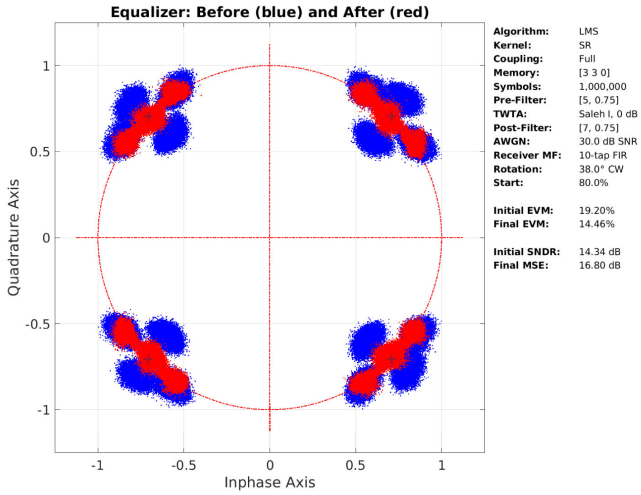


FIGURE 10. Trilobate patterns in equalized QPSK constellation, 30 dB SNR.

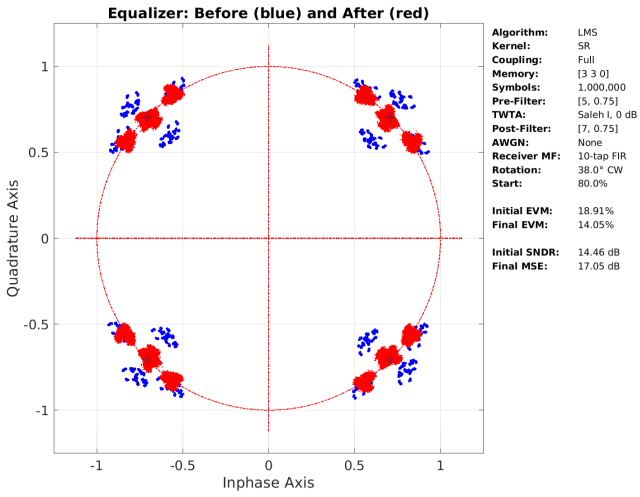


FIGURE 11. Receiver MF pattern with 1,000,000 symbols.

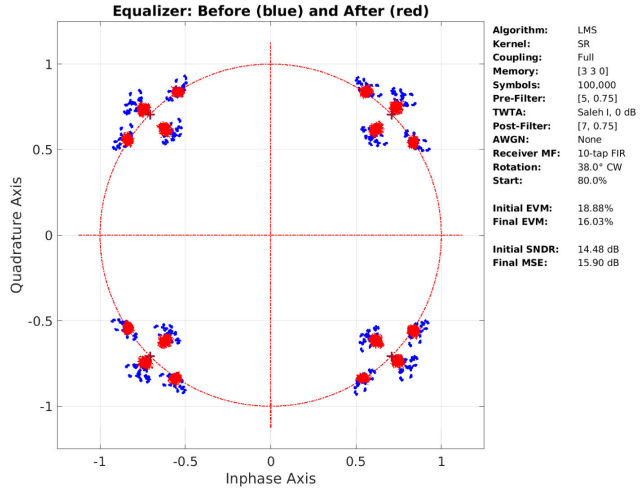


FIGURE 12. Receiver MF pattern with 100,000 symbols.

pre- and post-filters that controls the distribution of unequalized symbol points, while AWGN blurs this distribution. Moreover, it is largely *the receiver matched filter* that has the greatest influence on equalizer behavior, as it determines the *circumferential* distribution of symbols in the unequalized constellation.

B. EFFECTS OF THE RECEIVER MATCHED FILTER

Eliminating the receiver matched filter significantly changes the distribution of symbols in the unequalized (blue) constellation, as illustrated in Figure 13. The more uniform distribution of symbols allows for greatly improved equalizer performance (∼6.5 dB reduction in MSE), with significant equalization in both the radial and circumferential directions. Spread in the circumferential direction is considerably lower than that visible in Figure 12. This is empirical evidence that the receiver MF is not optimally matched to the nonlinear channel; the adaptive equalizer provides a better match.

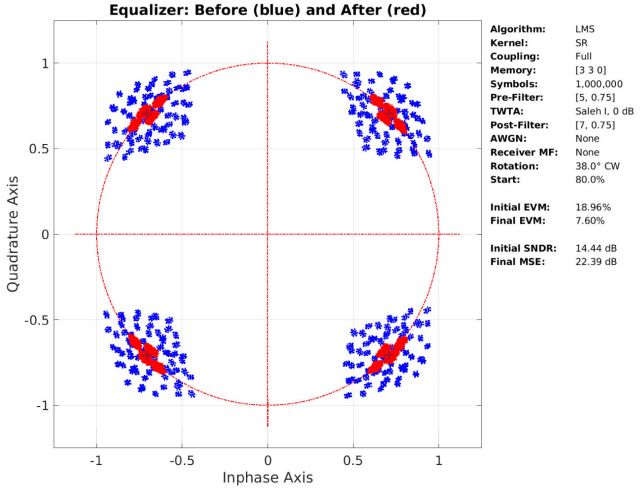


FIGURE 13. Improved radial convergence after 1,000,000 symbols.

C. COUPLING

Next we consider the effects of partially-decoupling the equalizer kernels, as was discussed in Section V-B. Comparing the equalized constellation in Figure 14, obtained using a partially-decoupled equalizer, to that shown previously in Figure 9, we see that partially-decoupling the kernels primarily affects equalization in the *radial* direction. Partial-decoupling significantly slows down the (already slow) slow radial convergence mode, most noticeably affecting the constellation points outside of the unit circle. As in the case of the fully-coupled equalizer, the constellation points nearest to the decision boundaries are largely unaffected. Therefore, partial decoupling has a relatively small effect (∼0.6 dB in this case) on MSE.

D. MEMORY LENGTH AND SIMPLIFIED KERNELS

In Section VI it was shown that kernel symmetry restriction and convolution factoring may be used to realize significant gains in computational efficiency. In this section, we show that choosing "simplified" kernels of the lowest memory length needed to minimize equalizer MSE will similarly

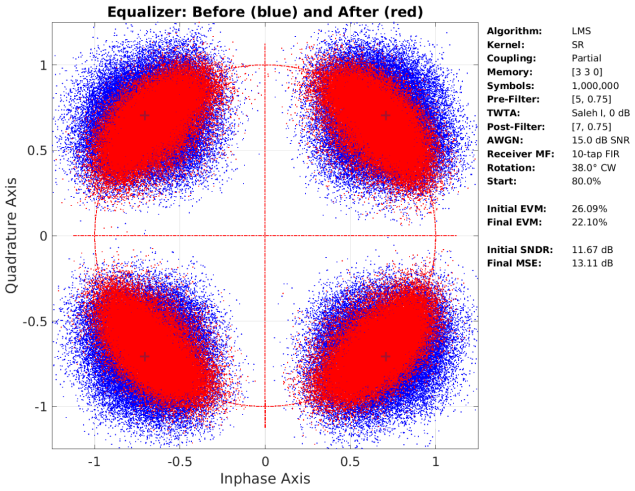


FIGURE 14. Partially-Decoupled adaptive equalizer, 15db SNR.

provide significant gains in computational efficiency. Counterintuitively, for most modulation schemes, adding memory to an adaptive LMS Volterra equalizer beyond a certain memory length will *not* decrease the LMS error of the filter. Use of simplified kernels was suggested by Gutierrez and Ryan [11]; here we extend their work by considering all combinations of linear and cubic memory length from [0, 0, 0] to [10, 10, 0] for BPSK, QPSK, and 8-PSK, and 16-QAM modulation, and all combinations of cubic and quintic memory from [10, 0, 0] to [10, 10, 10] for 16-QAM. We plot the MSE of the converged LMS equalizers in a series of heatmap grids.

E. HEATMAPS

Heatmaps, such as Figure 16, are presented for each modulation scheme. Linear memory length N_1 is shown on the horizontal axis, while cubic memory length N_3 is shown on the vertical axis. A memory length of zero indicates no memory; i.e., the respective kernel is not used in the equalizer. Values of linear and cubic memory length N_1 and N_3 range from 0 to 10. Converged equalizer MSE values in dB are shown in the heatmap, with color gradation scale on the right. Lower MSE indicates a higher degree of equalization.

The first *row* of each linear/cubic heatmap shows the MSE of a conventional LMS adaptive *linear* Volterra equalizer, a baseline for measuring cubic equalizer performance. It can be seen that including a cubic kernel generally improves equalization by ∼2 dB compared to a linear kernel alone. The first *column* of the linear/cubic heatmap shows the MSE for equalizers which lack a linear kernel; these *Volterra-like* equalizers generally perform poorly.

F. MODULATION TYPES

Each modulation scheme (BPSK, QPSK, 8-PSK, and 16-QAM) is considered separately in the subsections that follow. A representative scatter plot showing equalizer convergence is presented for each type of modulation, followed by two heatmaps: the first for the fully-coupled LMS adaptive equalizer, the second for the partially-decoupled equivalent. Optimum combinations of memory length, if apparent in the heatmap, are discussed. In the case of 16-QAM, additional heatmaps of quintic vs. cubic memory (with linear memory held constant) are also provided.

1) BPSK

BPSK is more immune to noise than other M-PSK modulation schemes because of its wide symbol separation. For a distorted but noise-free BPSK signal, the LNL filter cascade causes rotation, with bifurcation occurring mainly in the radial direction, where the adaptive Volterra equalizer works best. For this reason adaptive Volterra equalizers of various memory length are very effective in equalizing BPSK. Figure 15 shows the action of an equalizer of memory length $N = [3, 3, 0]$ on a distorted BPSK constellation.

Heatmaps for fully-coupled and partially-decoupled LMS adaptive Volterra equalizers for BPSK are shown in Figures 16 and 17, respectively. The MSE values for equalizing BPSK are typically much lower than those for QPSK and for other modulation schemes, approaching -50 dB. Equalization levels off for linear memory length $N_1 \geqslant 4$. Adding cubic memory incrementally improves equalizer performance; there is no obvious optimal cubic memory length and N_3 must therefore be selected based on desired final MSE and tolerable computational burden. This is unusual, as the other modulation schemes analyzed herein have an optimal (or near-optimal) cubic memory value.

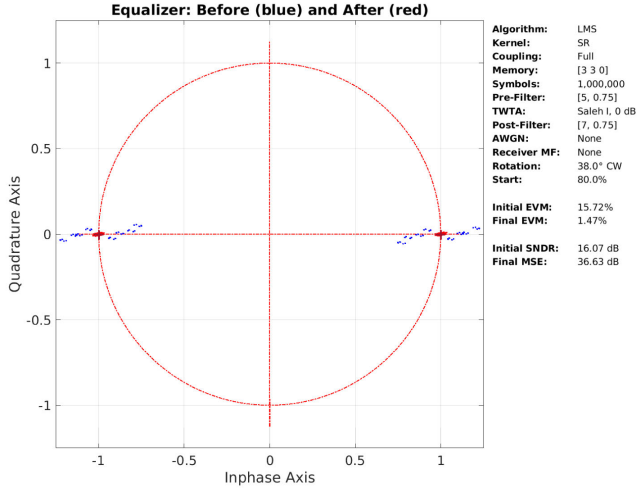


FIGURE 15. Equalized BPSK constellation; full-coupling, $N = [3, 3, 0]$.

2) QPSK

Figure 18 shows the effect of an equalizer of memory length $N = [10, 10, 0]$ operating on a distorted QPSK constellation; note the good radial convergence. Comparing this figure to Figure 13, in which $N = [3, 3, 0]$, we note that, counterintuitively, increasing the number of memory units had a detrimental effect on equalization.

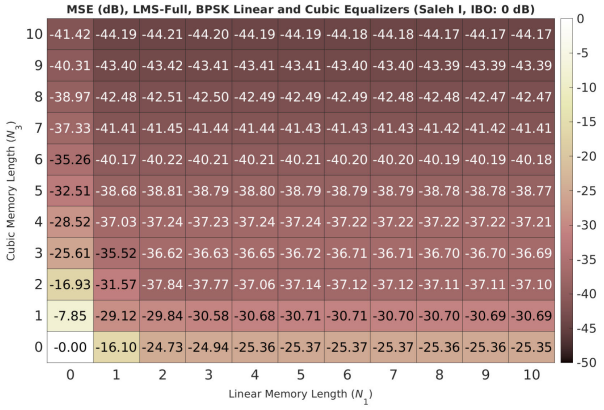


FIGURE 16. MSE, fully-coupled LMS Volterra equalizers, BPSK.

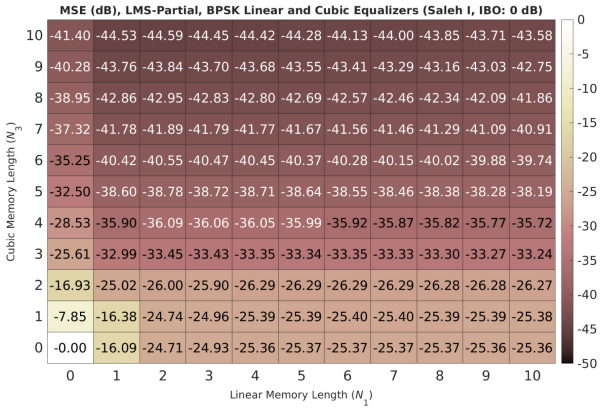


FIGURE 17. MSE, partially-decoupled LMS Volterra equalizers, BPSK.

Figures 19 and 20 are heatmaps for fully-coupled and partially-decoupled adaptive Volterra equalizers, respectively, for QPSK. It can be seen in these figures that optimal linear memory is $N_1 = 4$ and optimal cubic memory is $N_3 = 5$; increasing linear or cubic memory length beyond these values provides no significant improvement in equalization. For QPSK, there is little practical difference in MSE between the full-coupled and partial-decoupled LMS equalizers.

3) 8-PSK

The closer symbol spacing in 8-PSK makes it less immune to distortion and noise than BPSK or QPSK. Figure 21 shows the effect of an $N = [3, 3, 0]$ fully-coupled adaptive LMS equalizer on 8-PSK. The nonlinear distortion causes more uniform (and more dense) scatter than in the case of QPSK. Figures 22 and 23 are heatmaps for fully-coupled and partially-decoupled equalizers, respectively, for 8-PSK. As can be seen in the heatmaps, memory lengths reach the point of diminishing returns for $N_1 \geqslant 4$ and $N_3 \geqslant 4$. As expected, the fully-coupled equalizer performs slightly better than the partially-decoupled equalizer for 8-PSK.

4) 16-QAM

Equalizing a distorted QAM constellation is more challenging than equalizing the other modulation schemes considered heretofore. Some level of IBO is required for the transmission

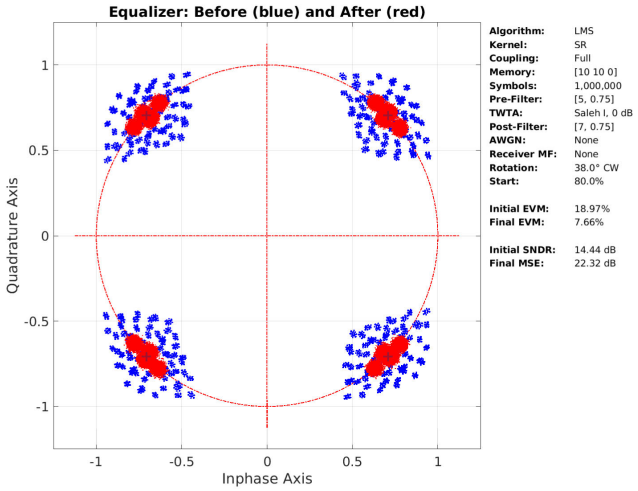


FIGURE 18. Equalization of nonlinear distortion; no MF $N = [10, 10, 10]$.

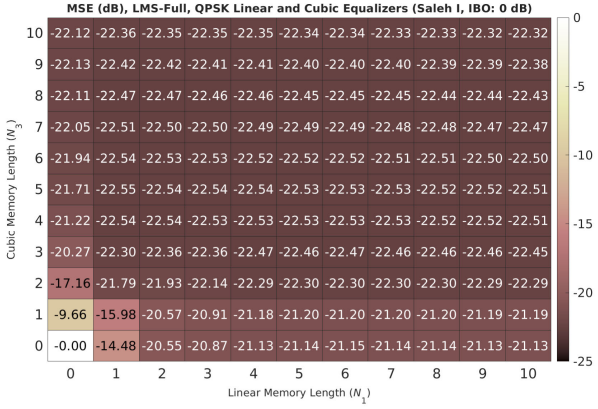


FIGURE 19. MSE for fully-coupled LMS Volterra equalizers, QPSK.

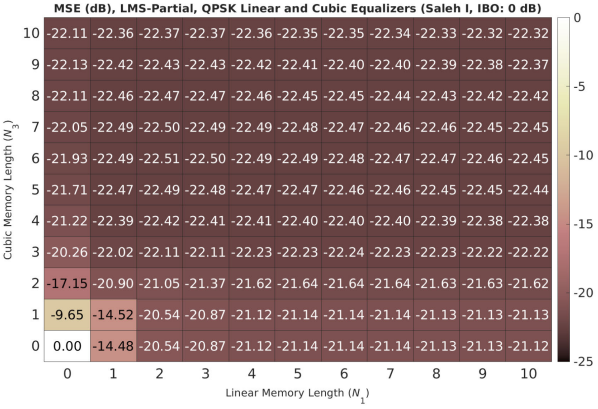


FIGURE 20. MSE for partially-decoupled LMS Volterra equalizers, QPSK.

of QAM using a TWTA, and a quintic kernel is needed for optimal Volterra equalization of QAM. While it is common practice to use square-root raised-cosine wave shapes for QAM, here we use rectangular pulse shapes, as a worst-case scenario. Both linear/cubic and linear/cubic/quintic equalizers were tested in equalizing 16-QAM with an IBO

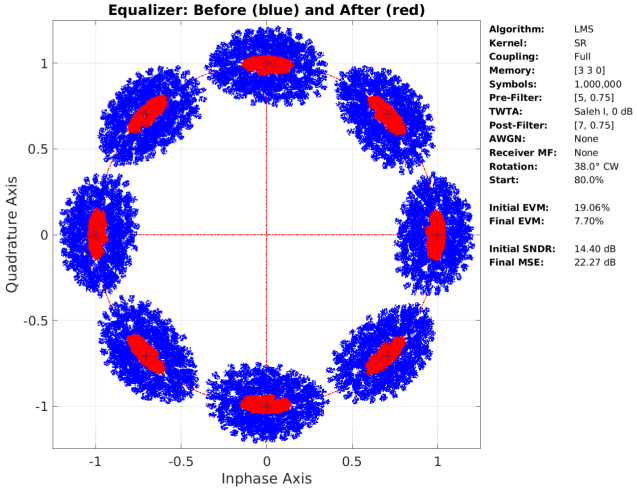


FIGURE 21. Equalized 8-PSK constellation; full-coupling, $N = [3, 3, 0]$.

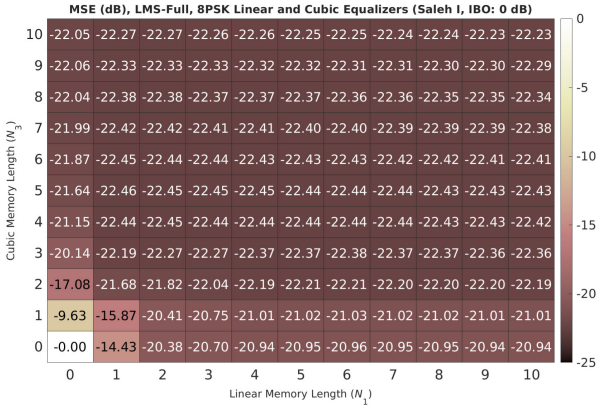


FIGURE 22. MSE, fully-coupled LMS Volterra equalizers, 8-PSK.

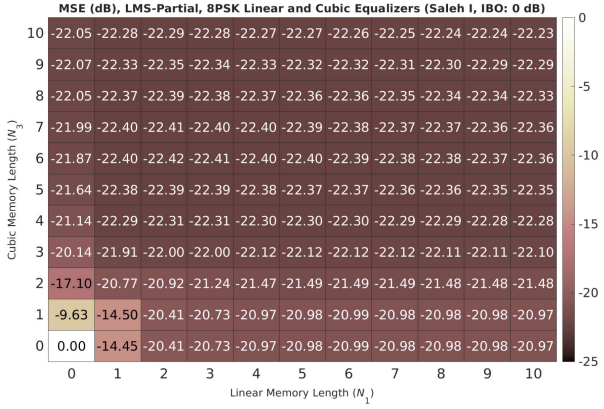


FIGURE 23. MSE, partially-decoupled LMS Volterra equalizers, 8-PSK.

of 6 dB. Both the fully-coupled and partially-decoupled LMS algorithms were used in the simulations. As with the other modulation schemes considered previously, LNL pre- and post-filters used were fifth- and seventh-order, respectively, with the *Saleh I* TWTA model and no receiver matched filter. However, artificial counter-rotation was *not*

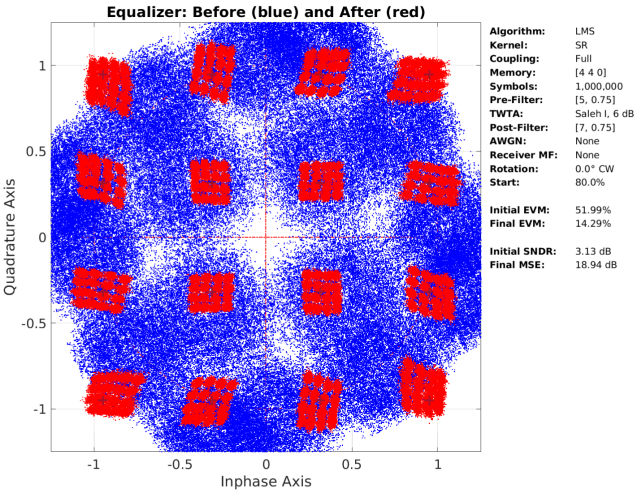


FIGURE 24. Equalized 16-QAM constellation; full-coupling, 6 dB IBO.

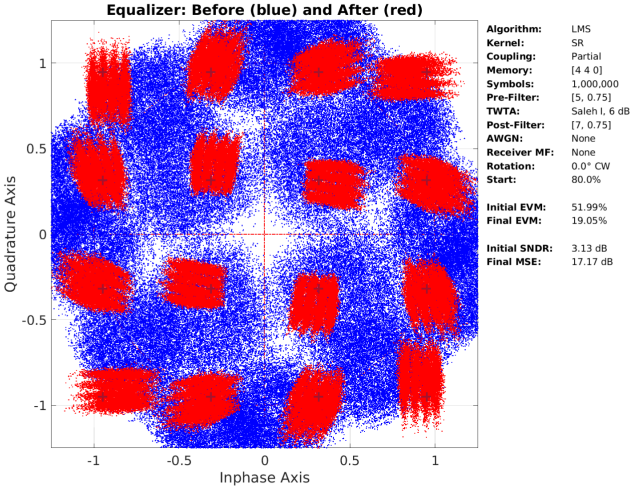


FIGURE 25. Equalized 16-QAM constellation; partial-decoupling, 6 dB IBO.

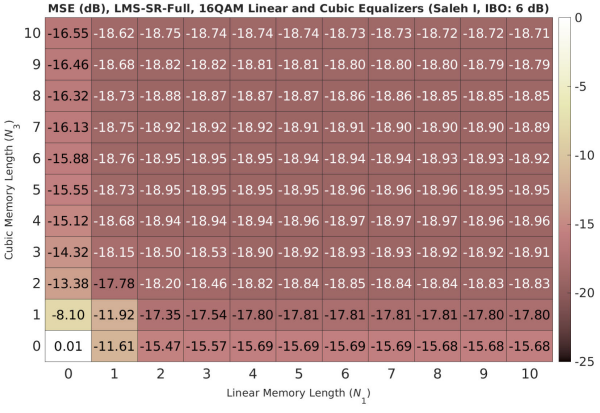


FIGURE 26. MSE, fully-coupled LMS Volterra equalizers, 16-QAM 6 dB IBO.

applied to the distorted 16-QAM constellations in the scatter plots, as alternative means of compensating for constellation warping (aside from IBO) are not generally used.

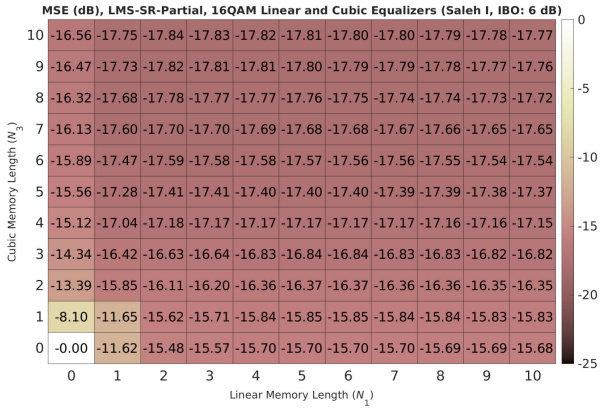


FIGURE 27. MSE, partially-decoupled LMS Volterra equalizers, 16-QAM 6 dB IBO vs. linear and cubic memory length.

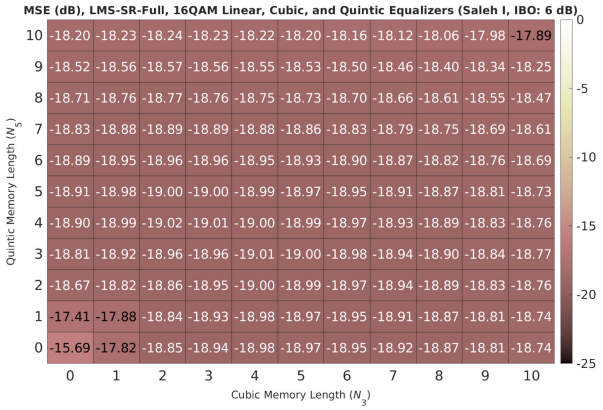


FIGURE 28. MSE, fully-coupled LMS quintic Volterra equalizers 16-QAM 6 dB IBO vs. cubic and quintic memory lengths.

Scatter plots for the fully-coupled and partially-decoupled cases with memory $N = [4, 4, 0]$ are shown in Figures 24 and 25, respectively, using an IBO of 6 dB. In each case, the linear/cubic equalizer worked well at handling the amplitude-dependent counter-rotation and producing tight groupings of the equalized symbols. Unsurprisingly, the equalizer performed slightly better (∼1dB) with IBO of 10 dB than 6 dB. The partially-decoupled equalizer produced a notable *streaking effect* in the equalized constellation, visible in Figure 25. The streaking is most prominent (and most problematic) at the corner points of the constellation. (In the extreme, as IBO approaches zero, the corner points of the equalized QAM constellation converge to the unit circle.)

Figures 26 and 27 are heatmaps for fully-coupled and partially-decoupled linear/cubic equalizers, respectively. It can be seen from these heatmaps that linear memory of $N_1 = 4$ and cubic memory of $N_3 = 3$ or $N_3 = 4$ provide the best equalization at the lowest computational burden. The partially-decoupled LMS equalizers underperformed the fully-coupled equalizers by ∼2 dB for 6 dB IBO. Given the streaking effect caused by the partially-decoupled equalizer, and the subsequent close proximity of the corner symbols to their adjacent symbols, partially-decoupled linear/cubic Volterra equalizers are not recommended for QAM. (Interestingly, for the fully-coupled case, the Volterra-like equalizers in the first column behave fundamentally differently than the full Volterra equalizers in the other columns; it is clear that adding even a single linear coefficient to these equalizers greatly improves performance, while adding a single cubic coefficient to the linear equalizers shown in the first row improves performance by ∼2 dB.)

Figure 28 shows the MSE for linear/cubic/quintic equalizers for 6 dB IBO, with a fixed linear memory length of $N_1 = 10$. Optimal MSE values occurred at or close to $N = [10, 4, 4]$. The MSE values in Figure 28 are very close to those in Figure 26, typically a fraction of a dB lower, at significantly greater computational burden. We can conclude that 16-QAM can be effectively equalized with a linear/cubic equalizer, as per [4]. Further simulation experiments using 3 dB IBO indicate that including a quintic kernel with memory length $N_5 \geqslant 4$ provided a slight reduction in MSE (∼0.1 dB to ∼0.5 dB over a liner/cubic equalizer; however, at this IBO the corner symbols are pulled fairly close to the unit circle, causing the constellation to resemble APSK more than QAM.) For IBO values greater than 3 dB, a linear/cubic equalizer performs practically as well as a linear/cubic/quintic equalizer at reduced computational burden.

IX. CONCLUSION

In this paper, we have examined the Saleh parameter sets found in the literature, analyzed two methods of increasing the computational efficiency of Volterra filters, and quantified the effects of channel memory and the receiver matched filter on LMS adaptive Volterra equalizer performance for common digital modulation schemes with TWTA distortion, after briefly considering the effects of AWGN. Further, we have compared the performance of fully-coupled and partially-decoupled adaptive Volterra equalizers for the various modulation schemes, and studied the effects of combinations of linear, cubic, and quintic memory length thereon.

A. RESULTS

The Saleh parameter sets found in the literature were collected and compared. They were found to differ mainly in their maximum (limiting) phase shift and may be classified as belonging to one of three groups (∼30°, ∼60°, and ∼110°); otherwise, all produce similar AM-AM distortion.

Very significant improvements in computational efficiency of the LMS adaptive Volterra equalizer were achieved by elimination of redundant coefficients by symmetry-restriction, convolution-sum factoring, and choice of simplified kernels. Exact polynomial expressions for computational efficiency were derived for several forms of the Volterra filter using these techniques alone and in combination; very significant reductions in computational burden were obtained, typically exceeding ∼90% and thereby making efficient FPGA implementation more feasible. Our simulation experiments show that simplified cubic and quintic kernels

produce equalization results practically identical to those of higher memory length.

In our simulations we have compared fully-coupled and partially-decoupled LMS adaptive Volterra filters in equalizing TWTA nonlinear distortion and found, as expected, that the partially-decoupled equalizers converge to values that are close to, but less than the optimal MSE values obtained by their fully-coupled counterparts.

Through simulation experiments we have shown that a receiver-based LMS adaptive Volterra equalizer can be effective in equalizing nonlinear TWTA distortion provided that channel memory is relatively low (i.e., the pre- and post-low-pass filters used in the LNL filter cascade are of order ∼6 or lower) so as not to produce a signal-to-distortion ratio (SDR) much lower than ∼15 dB. We have further shown that the adaptive Volterra equalizer alone provides a better match to the nonlinear channel than does a conventional receiver matched filter. A conventional receiver matched filter should be excluded entirely when an adaptive Volterra equalizer is used in the receiver, as it causes variance of the unequalized constellation in the circumferential direction in the I-Q plane. (Alternatively, the receiver MF could be replaced with a filter optimally matched to the distorted waveforms [45], or replaced with an adaptive pre-filter [11]).

Further, our simulation experiments have demonstrated that adaptive LMS equalizers tend to converge very slowly in the radial direction. Training with at least 10^5 symbols, and preferably 10^6 or more symbols, is needed for full convergence in the radial direction.

For the digital modulation schemes herein considered (BPSK, QPSK, 8-PSK, and 16-QAM), results of our simulation experiments indicate that a good rule-of-thumb is to use ''simplified'' linear and cubic Volterra filter kernels with four linear and four cubic memory units, as these appear to be optimal (or near-optimal) memory lengths. Using memory beyond four units provides little or no marginal improvement in equalizer performance, while increasing computational burden. For constant-modulus modulation schemes (BPSK, QPSK, 8-PSK), only linear and cubic Volterra kernels are needed; a quintic kernel provides no marginal improvement while greatly increasing computational burden. For modulation schemes with non-constant modulus such as M-QAM, the quintic kernel provides a modest (∼0.5 dB) additional reduction in MSE above that provided by the linear and cubic kernels provided that TWTA IBO is low (∼3 dB).

B. LIMITATIONS

We investigated two methods for efficient implementation of *any* Volterra filter: symmetry-restriction and convolution-sum factoring. As noted, these methods can be applied to any implementation of a Volterra filter to eliminate unnecessary redundant coefficients and multiplication operations. Tables 6 and 7 give polynomial expressions for the number of multiply and addition operations, respectively, incurred by the specified form Volterra filter for each iteration of the filter. These tables do not, however, include the overhead operations inherent in the particular language used to implement the Volterra filter. When used to implement an *adaptive* Volterra filter, the state vectors must be preserved to update the kernel coefficients during training. This will incur additional operations not counted in the polynomial expressions.

Partially-decoupled adaptive Volterra filters, while offering several distinct advantages (see Section V-B), including the ability to compute optimal step-size for each kernel order, are suboptimal and converge to a (usually) slightly higher MSE than their fully-coupled counterparts. Whether or not the higher error value is acceptable, or worth the speed obtained by using the optimal step-size, is an *engineering judgment* decision which may be of particular importance in implementing the equalizer on an FPGA. For PSK modulation, partial-decoupling may be used with a slight increase in MSE. It is clear, however, that partially-decoupled equalizes do not generally perform well in equalizing QAM.

C. FUTURE WORK

In this paper, we have only briefly considered the effects of AWGN on equalizer performance. Instead, we have focused our attention on more fundamental underlying issues: the effects of nonlinear distortion, channel memory, the receiver matched filter, and kernel memory length on the unequalized and equalized constellations. Having addressed these issues, proposed future work can now address the effects of AWGN on BER for the modulation schemes we considered.

Other areas of future research include investigating partially-decoupled LMS adaptive Volterra filters with optimal step-sizes for each kernel order, and further investigating the relationship between nonlinear channel memory and optimal equalizer memory. Another area of great interest to the authors is studying the effects of *individual* kernel coefficients, and possibly achieving additional gains in computational efficiency by removing entire *classes* of coefficients from the cubic and quintic kernels Volterra equalizers and investigating the resulting novel forms of the adaptive Volterra equalizer.

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JEROME J. MALONE JR. received the B.S. degree in engineering physics from Lehigh University, Bethlehem, PA, USA, in 1987, the M.S. degree in systems engineering from Colorado State University, Pueblo, CO, USA, in 2003, and the M.S. and Ph.D. degrees in electrical engineering from the University of Colorado at Colorado Springs, Colorado Springs, CO, USA, in 2007 and 2023, respectively.

Most recently, he held the position of Senior Systems Engineer with Union Pacific Railroad, Omaha, NE, USA, where he worked in positive train control (PTC) and RF propagation modeling in the Communications Systems Engineering Research and Development Department and the Telecommunications Engineering Department. Prior to that he held the position of Senior Engineer II at Transportation Technology Center Inc. (MxV Rail Inc.) in Pueblo, Colorado, USA, where he conducted research, development, and testing of various railroad-related positive train control components. He was responsible for managing various railroad-related interdisciplinary research and testing projects in the electrical, mechanical, and civil engineering fields. He has authored several articles on railroad-related topics and holds two patents for positive train control inventions. His research interests include circuit analysis, simulation of communication systems, applications of Volterra filters, railroad track circuits, moving-block systems, automatic detection of rail breaks, positive train control, and improved track geometry measurement techniques.

Dr. Malone is a Licensed Professional Engineer in the States of Nebraska and Colorado, USA. He holds the FCC GROL, RT(2C), GMDSS, and Amateur Extra Licenses. He was a recipient of the 2018 ASCE Wellington Prize.



BYEONG KIL LEE (Senior Member, IEEE) received the Ph.D. degree in computer engineering from The University of Texas, Austin, TX, USA, in 2005.

He is currently an Associate Professor with the Department of Electrical and Computer Engineering, University of Colorado at Colorado Springs. Prior to joining as a Faculty Member at UCCS, he worked at Samsung as the Vice President for five years. For four years, he was an Assistant Professor with The University of Texas at San Antonio (UTSA). For five years, he was a Senior Design Engineer with Texas Instruments (TI). For ten years, he was a Senior Research Staff with the Agency for Defense Development (ADD). His research interests include computer architecture, application-specific embedded systems (mobile processors), deep-learning-based intelligent computing, workload characterization of emerging applications, parallel computing and parallel architecture design, performance modeling, low-power design, and early-stage power estimation. His research has been published in several international conference proceedings and journals.

Dr. Lee serves on the Technical Program Committee and the Organizing Committee for some conferences and workshops, such as ISCA, ISPASS, IISWC, ICCD, HPCA, ASAP, PACT, ICPP, and UCAS. He serves as a reviewer for many conferences and journals.